



Started on	Thursday, 6 August 2026, 10:06 AM
State	Finished
Completed on	Thursday, 6 August 2026, 10:18 AM
Time taken	11 mins 33 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
```

```
{
```

```
    int i= 1;
```

```
    int s =1;
```

```
    while(s <= n)
```

```
    {
```

```
        i++;
```

```
        s += i;
```

```
    }
```

```
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

```

1  #include<stdio.h>
2
3  int main()
4  {
5      int n;
6      scanf("%d",&n);
7
8      int counter=0;
9      int i=1;
10     int s=1;
11
12     counter+=2;
13
14     while(1)
15     {
16         counter++;
17         if(!(s<=n))
18         {
19             break;
20         }
21
22         i++;
23         counter++;
24
25         s+=i;
26         counter++;
27     }
28
29     printf("%d",counter);
30
31 }
```

	Input	Expected	Got	
✓	9	12	12	✓
✓	4	9	9	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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[Kanishk P 251801127-2025-AIDS](#) ▾**K2****Started on** Thursday, 6 August 2026, 10:11 AM**State** Finished**Completed on** Thursday, 6 August 2026, 10:47 AM**Time taken** 35 mins 52 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
        printf("*");
    }
    else
    {
        for(int i=1; i<=n; i++)
        {
            for(int j=1; j<=n; j++)
            {
                printf("*");
                printf("*");
                break;
            }
        }
    }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer: (penalty regime: 0 %)

```
1  #include<stdio.h>
2  int main()
3  {
4      int n;
5      int count=0;
6      scanf("%d",&n);
7      count++;
8      if(n==1)
9      {
10         count++;
11     }
12
13     else
14     {
15         for(int i=1;i<=n;i++)
16         {
17             count++;
18             for(int j=1;j<=n;j++)
19             {
20                 count++;
21                 count++;
22                 count++;
23                 break;
24             }
25             count++;
26         }
27         count++;
28     }
```

```
29 | printf("%d",count);  
30 | }
```

	Input	Expected	Got	
✓	2	12	12	✓
✓	1000	5002	5002	✓
✓	143	717	717	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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Kanishk P 251801127-2025-AIDS ▾

K2

Started on	Thursday, 6 August 2026, 10:32 AM
State	Finished
Completed on	Thursday, 6 August 2026, 10:53 AM
Time taken	20 mins 50 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
    for (i = 1; i <= num; ++i)
    {
        if (num % i == 0)
        {
            printf("%d ", i);
        }
    }
}
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include<stdio.h>
2
3  int main()
4  {
5      int n;
6      int count=0;
7      scanf("%d",&n);
8      count++;
9      for(int i=1;i<=n;i++)
10     {
11         count++;
12
13         if(n%i==0)
14         {
15             count++;
16         }
17         count++;
18     }
19
20     printf("%d",count);
21 }
```

	Input	Expected	Got	
✓	12	31	31	✓
✓	25	54	54	✓
✓	4	12	12	✓

Passed all tests! 

Correct

Marks for this submission: 1.00/1.00.

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Started on	Thursday, 6 August 2026, 10:53 AM
State	Finished
Completed on	Thursday, 6 August 2026, 11:01 AM
Time taken	7 mins 18 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include<stdio.h>
2
3  int main()
4  {
5      int n;
6      scanf("%d",&n);
7
8      int count=0;
9
10     for(int i=n/2;i<n;i++)
11     {
12         count++;
13         for(int j=1;j<n;j=2*j)
14         {
15             count++;
16             for(int k=1;k<n;k=k*2)
17             {
18                 count++;
19                 count++;
20             }
21             count++;
22
23         }
24         count++;
25     }
26     count++;
27     count++;
28     printf("%d",count);
29 }
```

	Input	Expected	Got	
✓	4	30	30	✓

	Input	Expected	Got	
✓	10	212	212	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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Started on	Thursday, 6 August 2026, 11:01 AM
State	Finished
Completed on	Tuesday, 25 August 2026, 11:00 AM
Time taken	18 days 23 hours
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
    int rev = 0, remainder;
    while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n /= 10;
    }
    print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include<stdio.h>
2  int main()
3  {
4      int n;
5      scanf("%d",&n);
6
7      int count=2;
8
9      int rev=0,remainder;
10     while(n!=0)
11     {
12         count++;
13         remainder=n%10;
14         count++;
15         rev=rev*10+remainder;
16         count++;
17         n/=10;
18         count++;
19     }
20     count++;
21     printf("%d",count);
22 }
```

	Input	Expected	Got	
✓	12	11	11	✓
✓	1234	19	19	✓

Passed all tests! 

Correct

Marks for this submission: 1.00/1.00.

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